

AWS EC2 + Ansible Setup Guide

PART 1 — Create and Configure AWS EC2 Servers

1. Create 3 EC2 instances using Amazon Linux and t3.micro.

Names:
- Controller-Node
- Node1
- Node2

2. Connect servers using MobaXterm and install packages:

```
sudo yum update -y  
sudo yum install net-tools wget vim -y
```

3. Change Hostname:

```
sudo hostnamectl set-hostname Controller-Node  
sudo hostnamectl set-hostname Node1  
sudo hostnamectl set-hostname Node2
```

4. Configure /etc/hosts:

```
sudo vim /etc/hosts
```

Example:

```
172.31.24.135    Controller-Node  
172.31.30.120   Node1  
172.31.31.109   Node2
```

5. Create user in all servers:

```
sudo useradd narasimha  
sudo passwd narasimha
```

Password: 1234

6. Give sudo access:

```
sudo vim /etc/sudoers
```

Add:

```
root ALL=(ALL) ALL  
narasimha ALL=(ALL) NOPASSWD:ALL
```

7. Enable Password Authentication:

```
sudo vim /etc/ssh/sshd_config
```

Change:

```
PasswordAuthentication no
```

To:

```
PasswordAuthentication yes
```

8. Restart SSH:

```
sudo systemctl restart sshd
```

9. Switch user:

```
su - narasimha
```

10. Generate SSH Key:

```
ssh-keygen
```

11. Verify Keys:

```
ls ~/.ssh
```

Expected:

```
id_rsa
id_rsa.pub
```

PART 2 — SSH Connection Between Nodes

1. Connect to Node1 and Node2:

```
ssh narasimha@Node1
ssh narasimha@Node2
```

Password: 1234

2. Copy SSH Keys:

```
ssh-copy-id narasimha@Node1
ssh-copy-id narasimha@Node2
```

3. Test passwordless login:

```
ssh Node1
ssh Node2
```

PART 3 — Install and Configure Ansible

1. Install Ansible:

```
sudo dnf install ansible -y
```

2. Check Version:

```
ansible --version
```

3. Move to ansible directory:

```
cd /etc/ansible
```

4. Create hosts file:

```
sudo touch hosts
sudo chmod 777 hosts
```

5. Edit hosts file:

```
sudo vim hosts
```

Add:

```
[webservers]
Node1
Node2
```

```
[appservers]
app1
app2
app3
```

```
[dbservers]
db1
db2
```

```
host1
host2
```

6. Configure ansible.cfg:

```
sudo vim ~/.ansible.cfg
```

Add:

```
[defaults]
interpreter_python=auto_silent
```

7. Run Ansible command:

```
ansible -m command -a "uptime" webservers
```

8. Test ping:

```
ansible all -m ping
```

Troubleshooting

1. Check hosts file:

```
cat /etc/hosts
```

2. Check SSH status:

```
sudo systemctl status sshd
```

3. Install openssh-clients if ssh-copy-id fails:

```
sudo yum install openssh-clients -y
```

4. Test SSH manually:

```
ssh Node1
```

```
ssh Node2
```